

## NETWORK LOAD BALANCER (NLB)

A Network Load Balancer (NLB) in Amazon Web Services is a load balancing service that distributes incoming network traffic across multiple servers (targets) using Layer 4 (Transport Layer) protocols like TCP, UDP, and TLS.

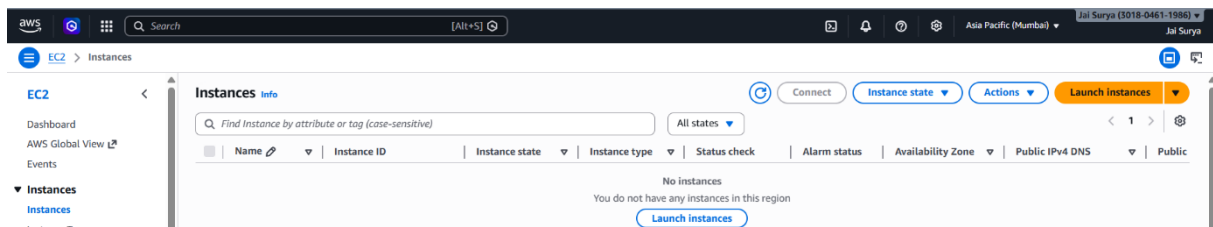
### Key Points

- Works at **Layer 4 (Transport Layer)**
- Handles **millions of requests per second**
- Provides **ultra-low latency**
- Supports **static IP addresses**
- Can handle **TCP, UDP, TLS traffic**
- Ideal for **high-performance and real-time applications**

## HANDS-ON FOR NETWORK LOAD BALANCER

### Screenshot: Create EC2 Instance

- Go to EC2 Dashboard
- Click on Create Instance



### Screenshot: Configure Instance Details

- Enter the Instance Name
- Set the Number of Instances to 2
- Select Amazon Linux as the OS
- Choose an existing Key Pair or create a new one

- aws

Search

[Alt+S]

EC2

Instances

Launch an instance

Asia Pacific (Mumbai)

Jai Surya (2018-0461-1396)

Jai Surya

Launch an instance

Amazon EC2 allows you to create virtual machines, or instances, that run on the AWS Cloud. Quickly get started by following the simple steps below.

Name and tags

Name

linux

Add additional tags

Application and OS Images (Amazon Machine Image)

An AMI contains the operating system, application server, and applications for your instance. If you don't see a suitable AMI below, use the search field or choose [Browse more AMIs](#).

Search our full catalog including 1000s of application and OS images

Recents

My AMIs

Quick Start

Amazon Linux

macOS

Ubuntu

Windows

Red Hat

SUSE Linux

Debian

Browse more AMIs

Including AMIs from AWS, Marketplace and the Community

Amazon Machine Image (AMI)

Summary

Number of instances

2

When launching more than 1 instance, consider EC2 Auto Scaling

Software Image (AMI)

Amazon Linux 2023 AMI 2023.10...read more

ami-0F559c3642608c138

Virtual server type (instance type)

t3.micro

Firewall (security group)

New security group

Storage (volumes)

1 volume(s) - 8 GiB

Cancel

Launch instance

Preview code

Key pair (login)

You can use a key pair to securely connect to your instance. Ensure that you have access to the selected key pair before you launch the instance.

Key pair name - required

3pmbatch2

Create new key pair

## Screenshot: Connect to EC2 Instances

- Select the first EC2 instance
- Click on Connect
- Use EC2 Instance Connect / SSH to connect
- Repeat the same steps to connect to the second EC2 instance



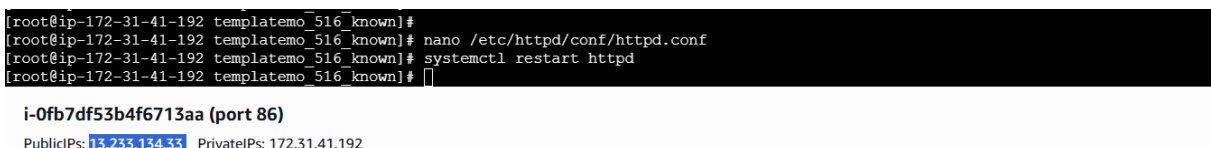
## Screenshot: Execute HTTPD Web Hosting Commands

- In the first instance Linux terminal, execute HTTPD web hosting commands



## Screenshot: Edit HTTPD Configuration File

- Execute nano /etc/httpd/conf/httpd.conf in the terminal



### Screenshot: Modify Listen Port

- Navigate to the Listen 80 line in the configuration file

```

GNU nano 8.3 /etc/conf/httpd.conf
# ServerRoot is a non-local disk, be sure to specify a local disk on the
# Mutex directive, if file-based mutexes are used. If you wish to share the
# same ServerRoot for multiple httpd daemons, you will need to change at
# least PidFile.
#
ServerRoot "/etc/httpd"
#
#
# Listen: Allows you to bind Apache to specific IP addresses and/or
# ports, instead of the default. See also the <VirtualHost>
# directive.
#
# Change this to Listen on a specific IP address, but note that if
# httpd.service is enabled to run at boot time, the address may not be
# available when the service starts. See the httpd.service(8) man
# page for more information.
#
#Listen 12.34.56.78:80
Listen 80
#
#
# Dynamic Shared Object (DSO) Support
#
# To be able to use the functionality of a module which was built as a DSO you
# have to place corresponding 'LoadModule' lines at this location so the
# directives contained in it are actually available before they are used.
#
# Statically compiled modules (those listed by 'httpd -l') do not need
#
^G Help          ^O Write Out    ^G Where Is     ^K Cut          ^Z Execute      ^C Location     M-U Undo        M-R Set Mark    M-I To Bracket
^X Exit          ^R Read File    ^W Replace      ^U Paste        ^J Justify      ^V Go To Line    M-E Redo        M-C Copy        M-? Where Was

```

### Screenshot: Save Configuration Changes

- Change Listen 80 to Listen 86
- Press Ctrl + X, then press Y and Enter to save changes

[illegible]

### Screenshot: Restart Service and Access Instance

- Execute `systemctl restart httpd`
- Copy the Public IP address of the instance

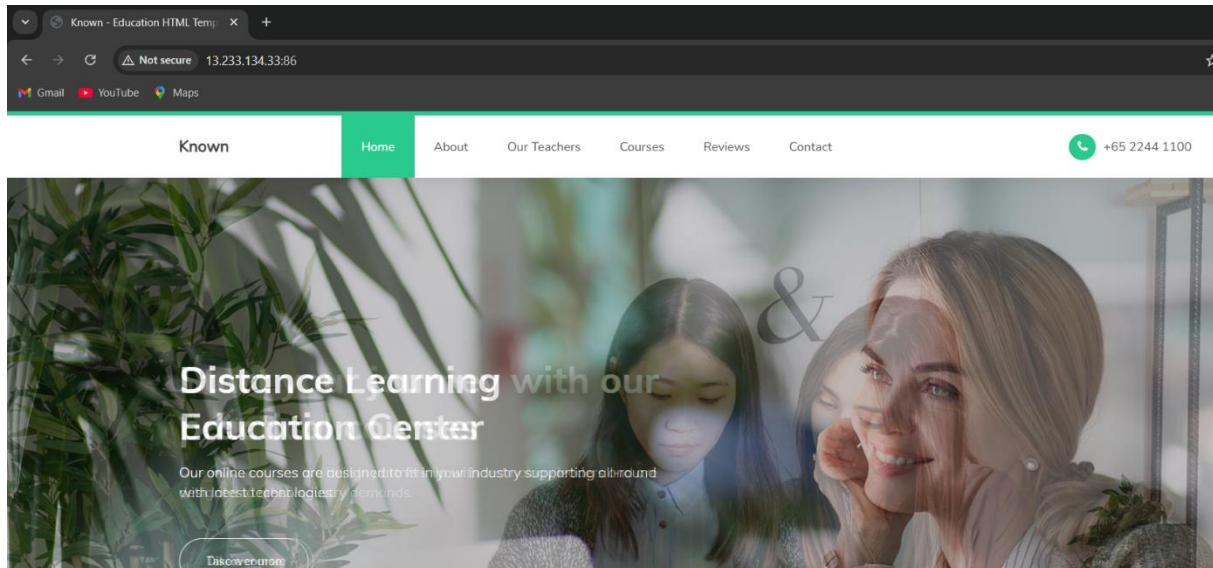
```
[root@ip-172-31-41-192 templatememo 516~known]#  
[root@ip-172-31-41-192 templatememo 516~known]# nano /etc/httpd/conf/httpd.conf  
[root@ip-172-31-41-192 templatememo 516~known]# systemctl restart httpd  
[root@ip-172-31-41-192 templatememo 516~known]#
```

**i-0fb7df53b4f6713aa (port 86)**

PublicIPs: **13.233.134.33** PrivateIPs: 172.31.41.192

## Screenshot: Access Website on Custom Port

- Paste the Public IP address in a new browser tab
- Add :86 at the end of the URL (Example: http://<Public-IP>:86)
- Verify that the website opens successfully



## Screenshot: Execute HTTPD Web Hosting Commands

- In the second instance Linux terminal, execute HTTPD web hosting commands

```
aws [Search] [Alt+S] Asia Pacific (Mumbai) Jai Surya (3018-0461-1986) Jai Surya
inflating: templatemo_516_known/js/jquery.js
inflating: templatemo_516_known/js/owl.carousel.min.js
inflating: templatemo_516_known/js/smoothscroll.js
[root@ip-172-31-38-78 ec2-user]# ls
templatemo_516_known templatemo_516_known.zip
[root@ip-172-31-38-78 ec2-user]# cd templatemo_516_known/
[root@ip-172-31-38-78 templatemo_516_known]# ls
css fonts images index.html js
[root@ip-172-31-38-78 templatemo_516_known]# mv * /var/www/html/
[root@ip-172-31-38-78 templatemo_516_known]# ls
[root@ip-172-31-38-78 templatemo_516_known]# cd /var/www/html/
[root@ip-172-31-38-78 html]# ls
css fonts images index.html js
[root@ip-172-31-38-78 html]# history
1  yum update -y
2  yum install httpd
3  systemctl status httpd
4  systemctl start httpd
5  systemctl status httpd
6  wget -S --content-disposition https://templatemo.com/download/templatemo_516_known
7  ls
8  unzip templatemo_516_known.zip
9  ls
10 cd templatemo_516_known/
11 ls
12 mv * /var/www/html/
13 ls
14 cd /var/www/html/
15 ls
16 history
[root@ip-172-31-38-78 html]#
```

i-07263d90ecd3ef8a (linux-77)  
PublicIPs: 65.1.131.199 PrivateIPs: 172.31.38.78

## Screenshot: Edit HTTPD Configuration File

- Execute nano /etc/httpd/conf/httpd.conf in the terminal
- Navigate to the Listen 92 line in the configuration file
- Change Listen 80 to Listen 92
- Press Ctrl + X, then press Y and Enter to save changes

```
GNU nano 8.3 /etc/httpd/conf
# ServerRoot: The top of the directory tree under which the server's
# configuration, error, and log files are kept.
#
# Do not add a slash at the end of the directory path.  If you point
# ServerRoot at a non-local disk, be sure to specify a local disk on the
# Mutex directive, if file-based mutexes are used.  If you wish to share the
# same ServerRoot for multiple httpd daemons, you will need to change at
# least PidFile.
#
ServerRoot "/etc/httpd"

#
# Listen: Allows you to bind Apache to specific IP addresses and/or
# ports, instead of the default.  See also the <VirtualHost>
# directive.
#
# Change this to Listen on a specific IP address, but note that if
# httpd.service is enabled to run at boot time, the address may not be
# available when the service starts.  See the httpd.service(8) man
# page for more information.
#
#Listen 12.34.56.78:92
Listen 92

#
# Dynamic Shared Object (DSO) Support
#

^G Help      ^O Write Out  ^F Where Is   ^K Cut        ^T Execute    ^C Location
^X Exit      ^R Read File  ^\ Replace    ^U Paste      ^J Justify    ^_ Go To Li

i-05d37d8dd8c539d65 (port 92)
PublicIPs: 13.233.97.251  PrivateIPs: 172.31.41.219
```

## Screenshot: Restart Service and Access Instance

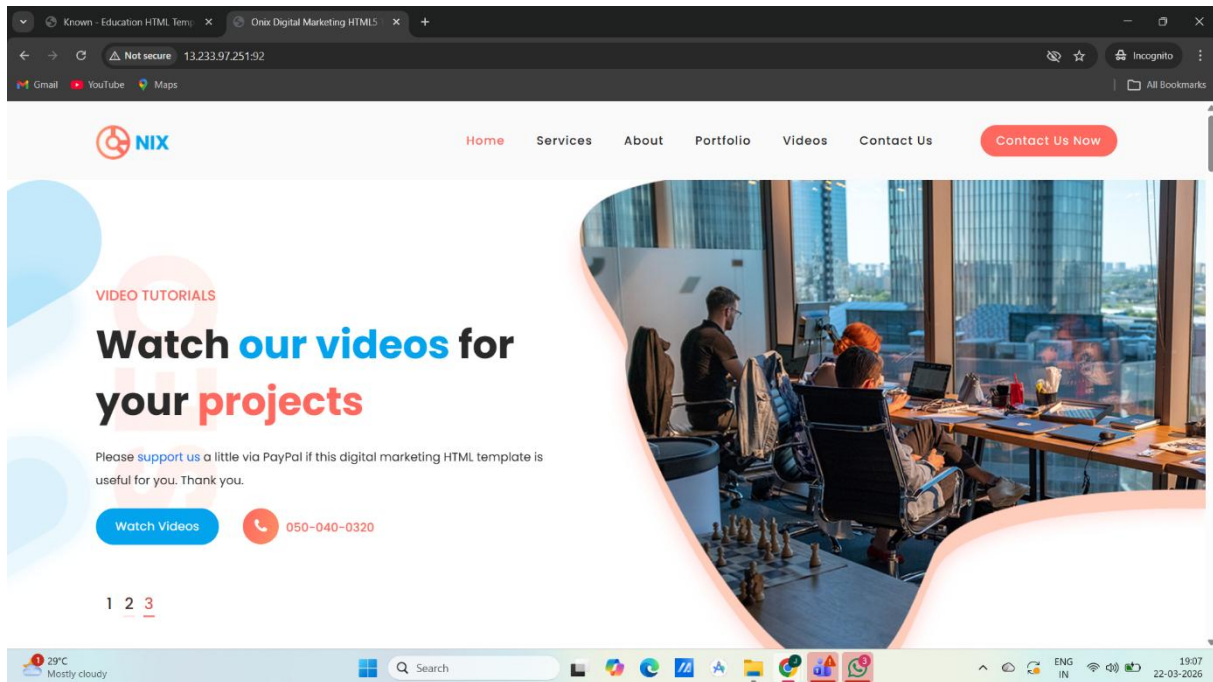
- Execute `systemctl restart httpd`
- Copy the Public IP address of the instance

```
[root@ip-172-31-41-192 templatememo 516 known]#
[root@ip-172-31-41-192 templatememo 516 known]# nano /etc/httpd/conf/httpd.conf
[root@ip-172-31-41-192 templatememo 516 known]# systemctl restart httpd
[root@ip-172-31-41-192 templatememo 516 known]#

i-0fb7df53b4f6713aa (port 86)
PublicIPs: 13.233.134.33  PrivateIPs: 172.31.41.192
```

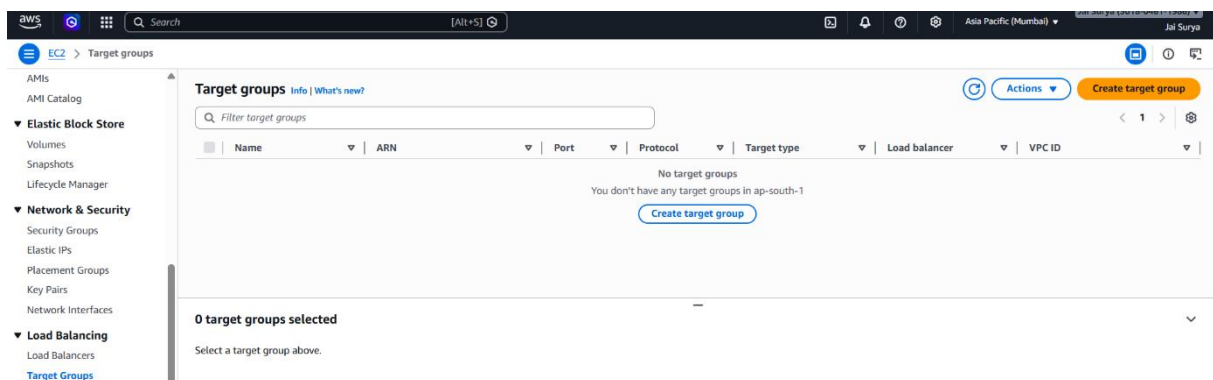
## Screenshot: Access Website on Custom Port

- Paste the Public IP address in a new browser tab
- Add :92 at the end of the URL (Example: `http://<Public-IP>:92`)
- Verify that the website opens successfully



## Screenshot: Create Target Group

- Go to Load Balancing section in EC2 Dashboard
- Click on Target Groups
- Click on Create target group



## Screenshot: Configure Target Group Settings

- Select Instances as the target type
- Enter the Target Group Name
- Set Protocol to TCP
- Set Port to 86



aws [Search] [Alt+S] Asia Pacific (Mumbai) Jai Surya 130%

EC2 > Target groups > Create target group

### Create target group

A target group can be made up of one or more targets. Your load balancer routes requests to the targets in a target group and performs health checks on the targets.

**Settings - immutable**  
Choose a target type and the load balancer and listener will route traffic to your target. These settings can't be modified after target group creation.

**Target type**  
Indicate what resource type you want to target. Only the selected resource type can be registered to this target group.

☒ **Instances**  
Supports load balancing to instances in a VPC. Integrate with Auto Scaling Groups or ECS services for automatic management.  
Suitable for: ALB NLB GWLB

☐ **IP addresses**  
Supports load balancing to VPC and on-premises resources. Facilitates routing to IP addresses and network interfaces on the same instance. Supports IPv6 targets.  
Suitable for: ALB NLB GWLB

☐ **Lambda function**  
Supports load balancing to a single Lambda function. ALB required as traffic source.  
Suitable for: ALB

☐ **Application Load Balancer**  
Allows use of static IP addresses and PrivateLink with an Application Load Balancer. NLB required as traffic source.  
Suitable for: NLB

**Target group name**  
Name must be unique per Region per AWS account.  
tg-86  
Accepts: a-z, A-Z, 0-9, and hyphen (-). Can't begin or end with hyphen. 1-32 total characters; Count: 5/32

**Protocol**  
Protocol for communication between the load balancer and targets.  
TCP

**Port**  
Port number where targets receive traffic. Can be overridden for individual targets during registration.  
86  
1-65535

## Screenshot: Configure Health Check

- Set Health Check Protocol to TCP
- Click on Next

**Health checks**  
The associated load balancer periodically sends requests, per the settings below, to the registered targets to test their status.

**Health check protocol**  
TCP

► **Advanced health check settings**

**Attributes**  
① Certain default attributes will be applied to your target group. You can view and edit them after creating the target group.

► **Tags - optional**  
Consider adding tags to your target group. Tags enable you to categorize your AWS resources so you can more easily manage them.

Cancel Next

## Screenshot: Register Targets

- Select the EC2 instance running on Port 86
- Click on Include as pending below

AWS Console Home Target groups > Create target group

Step 1: Create target group  
Step 2 - recommended: Register targets  
Step 3: Review and create

### Register targets - recommended

This is an optional step to create a target group. However, to ensure that your load balancer routes traffic to this target group you must register your targets.

**Available instances (1/2)**

Filter instances

|                                     | Instance ID         | Name    | State   | Security groups  | Zone        | Private IP  |
|-------------------------------------|---------------------|---------|---------|------------------|-------------|-------------|
| <input checked="" type="checkbox"/> | i-0fb7df53b4f6713aa | port 86 | Running | launch-wizard-13 | ap-south-1a | 172.31.41.1 |
| <input type="checkbox"/>            | i-05d37d8dd8c539d65 | port 92 | Running | launch-wizard-13 | ap-south-1a | 172.31.41.2 |

1 selected

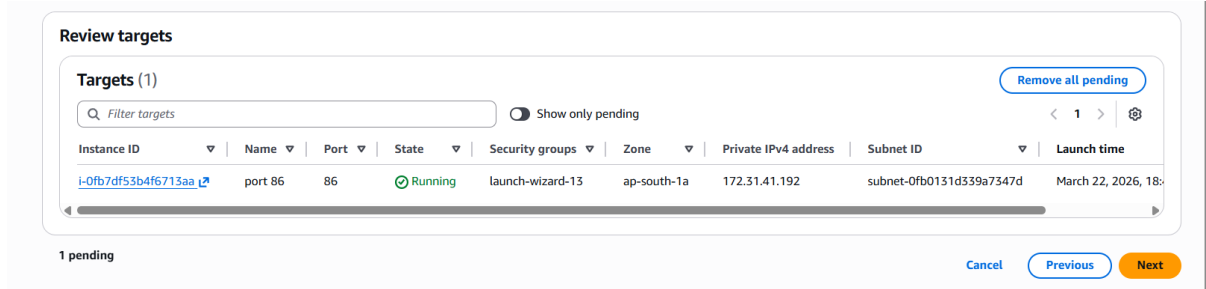
**Ports for the selected instances**  
Ports for routing traffic to the selected instances.  
86  
1-65535 (separate multiple ports with commas)

Include as pending below



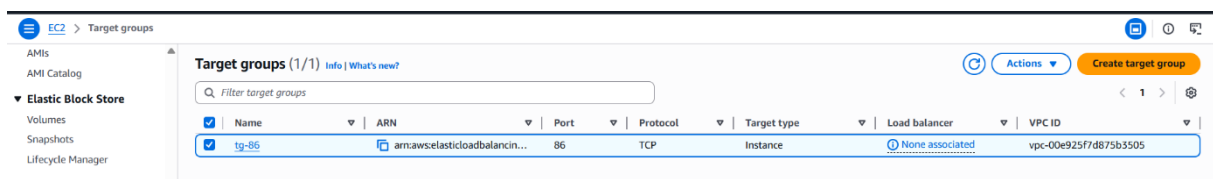
## Screenshot: Review and Create Target Group

- Click on Next
- Review the configuration
- Click on Create target group



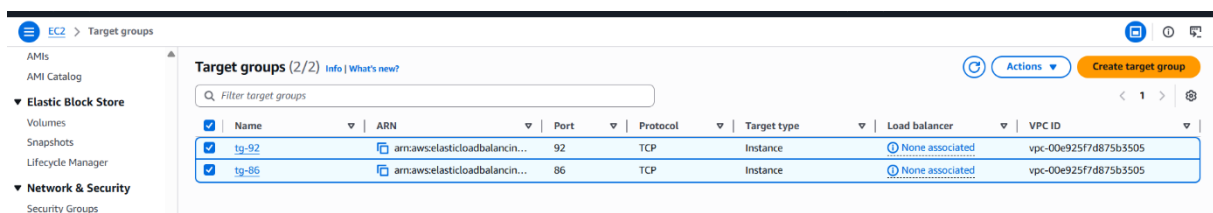
## Screenshot: Target Group Created (Port 86)

- Verify that the Target Group for Port 86 is created successfully
- Ensure the registered instance is listed and healthy



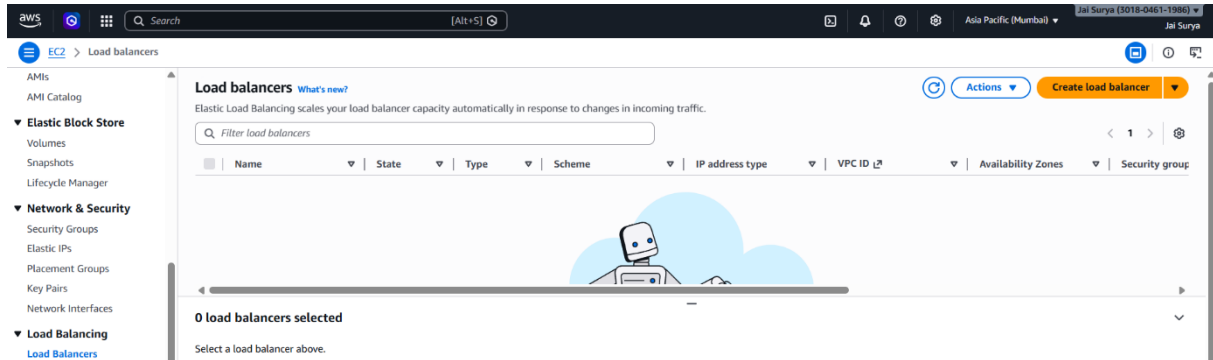
## Screenshot: Target Groups Created Successfully

- Repeat the same steps to create another target group for Port 92
- Verify that both target groups (Port 86 and Port 92) are created successfully



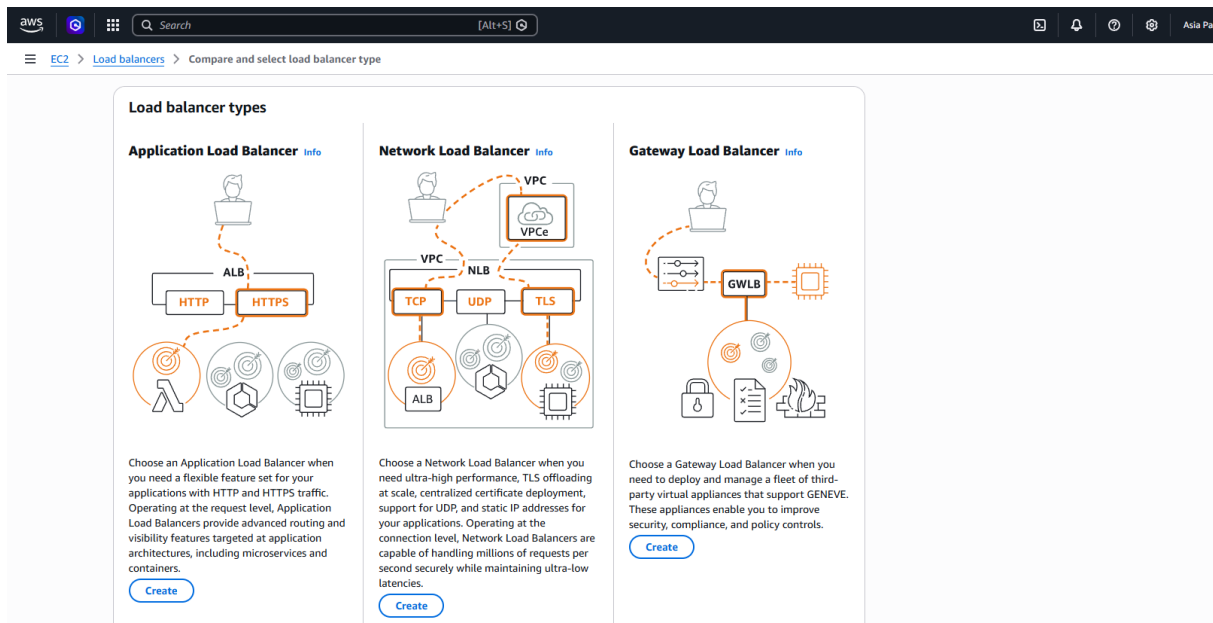
## Screenshot: Create Load Balancer

- Go to Load Balancers in EC2 Dashboard
- Click on Create Load Balancer



## Screenshot: Select Load Balancer Type

- Select Network Load Balancer



## Screenshot: Configure Load Balancer Basic Settings

- Enter the Load Balancer Name
- Select Internet-facing as the scheme

aws [Search] [Alt+S]

EC2 > Load balancers > Create Network Load Balancer

### Create Network Load Balancer [Info](#)

The Network Load Balancer distributes incoming TCP and UDP traffic across multiple targets such as Amazon EC2 instances, microservices, and containers. When the load balancer receives a connection request, it selects a port that are specified in the listener configuration, and the routing rule specified as the default action.

► How Network Load Balancers work

#### Basic configuration

**Load balancer name**  
Name must be unique within your AWS account and can't be changed after the load balancer is created.

network-lb

A maximum of 32 alphanumeric characters including hyphens are allowed, but the name must not begin or end with a hyphen.

**Scheme**  
Scheme can't be changed after the load balancer is created.

☒ Internet-facing

- Serves internet-facing traffic.
- Has public IP addresses.
- DNS name resolves to public IPs.
- Requires a public subnet.

☐ Internal

- Serves internal traffic.
- Has private IP addresses.
- DNS name resolves to private IPs.

**Load balancer IP address type** [Info](#)  
Select the front-end IP address type to assign to the load balancer. The VPC and subnets mapped to this load balancer must include the selected IP address types.

☒ IPv4

Includes only IPv4 addresses.

☐ Dualstack

Includes IPv4 and IPv6 addresses.

## Screenshot: Select Availability Zones and Subnets

- Select all Availability Zones (AZs)
- Choose the corresponding Subnets

aws [Search] [Alt+S] Asia Pacific (Mumbai) Jai Surya

EC2 > Load balancers > Create Network Load Balancer

### Network mapping [Info](#)

The load balancer routes traffic to targets in the selected subnets, and in accordance with your IP address settings.

**VPC**  
The load balancer will exist and scale within the selected VPC. The selected VPC is also where the load balancer targets must be hosted unless routing to on-premises targets or if using VPC peering. To confirm the VPC for your targets, view [target groups](#).

vpc-00e925f7d875b3505  
172.31.0.0/16 (default) [Create VPC](#)

**Availability Zones and subnets**  
Select one or more Availability Zones and corresponding subnets. Enabling multiple Availability Zones increases the fault tolerance of your applications. The load balancer routes traffic to targets in the selected Availability Zones only. Availability Zones that are not supported by the load balancer or the VPC are not available for selection.

☒ ap-south-1a (aps1-az1)

**Subnet**  
Only CIDR blocks corresponding to the load balancer IP address type are used. At least 8 available IP addresses are required for your load balancer to scale efficiently.

subnet-0fb0131d339a7347d  
172.31.32.0/20

**IPv4 address**  
The front-end IPv4 address of the load balancer in the selected Availability Zone.

☒ Assigned by AWS

☐ Use an Elastic IP address

☒ ap-south-1b (aps1-az3)

**Subnet**  
Only CIDR blocks corresponding to the load balancer IP address type are used. At least 8 available IP addresses are required for your load balancer to scale efficiently.

subnet-02af623bce16f5af0  
172.31.0.0/20

subnet-02af623bce16f5af0

**IPv4 address**  
The front-end IPv4 address of the load balancer in the selected Availability Zone.

☒ Assigned by AWS

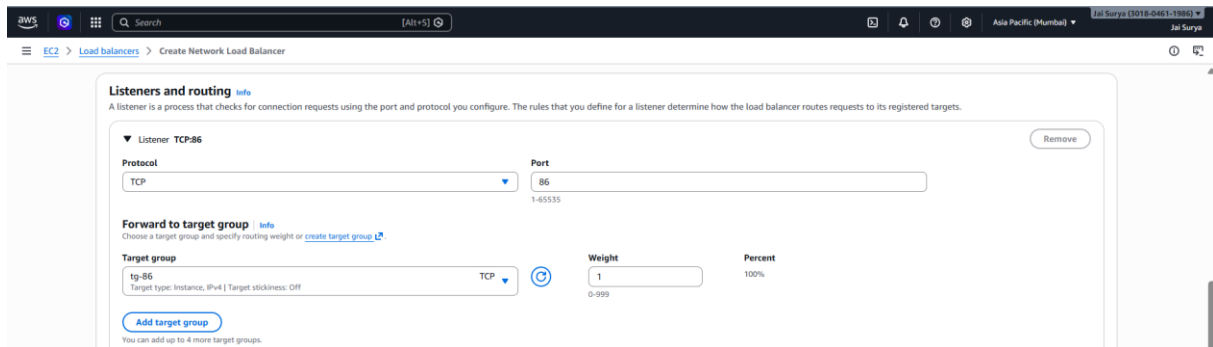
☐ Use an Elastic IP address

☒ ap-south-1c (aps1-az2)

**Subnet**  
Only CIDR blocks corresponding to the load balancer IP address type are used. At least 8 available IP addresses are required for your load balancer to scale efficiently.

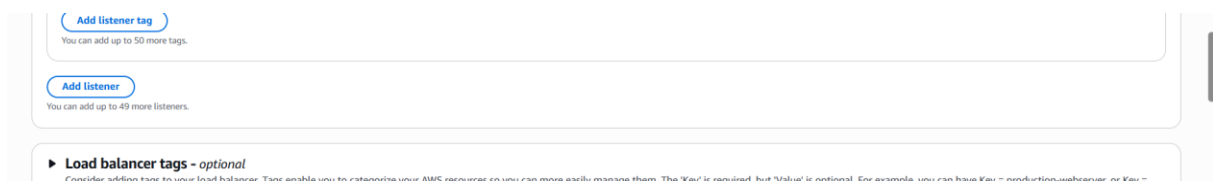
## Screenshot: Configure Listener and Routing

- Set Protocol to TCP
- Set Port to 86
- Select the Target Group (Port 86) for routing



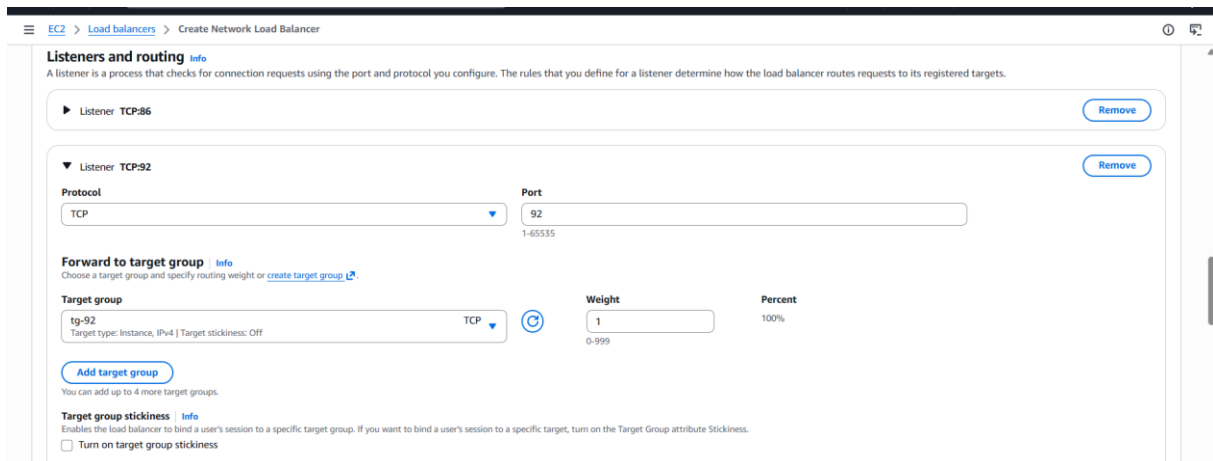
## Screenshot: Add Listener

- Click on Add listener



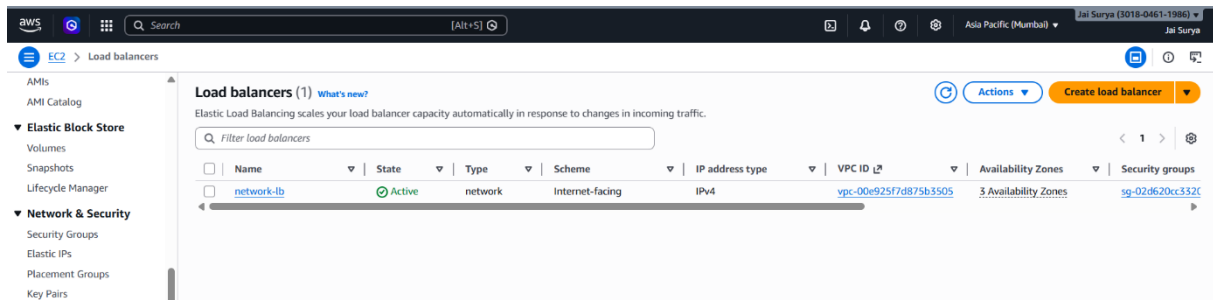
## Screenshot: Configure Second Listener

- Set Protocol to TCP
- Set Port to 92
- Select the Target Group (Port 92) for routing



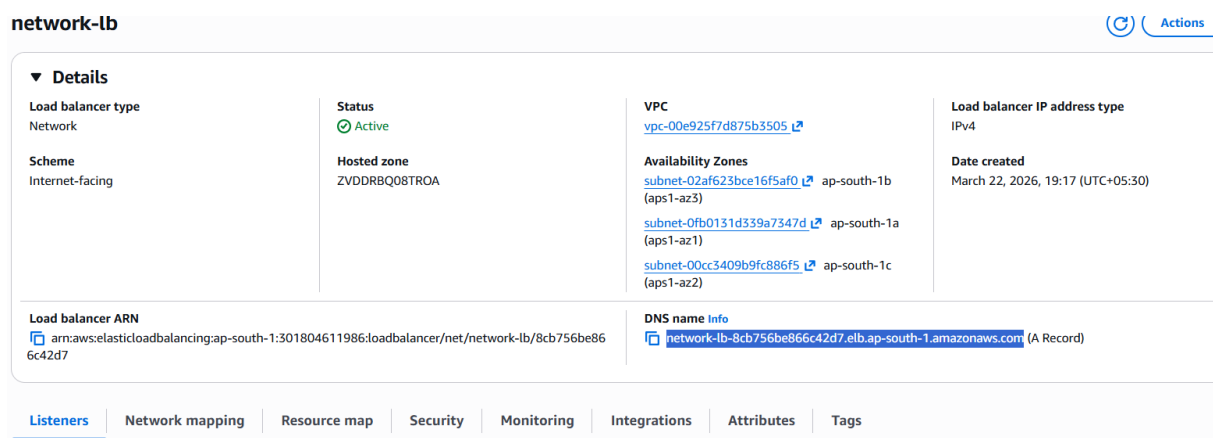
## Screenshot: Network Load Balancer Created

- Verify that the Network Load Balancer is created successfully
- Ensure both listeners (Port 86 and Port 92) are configured properly



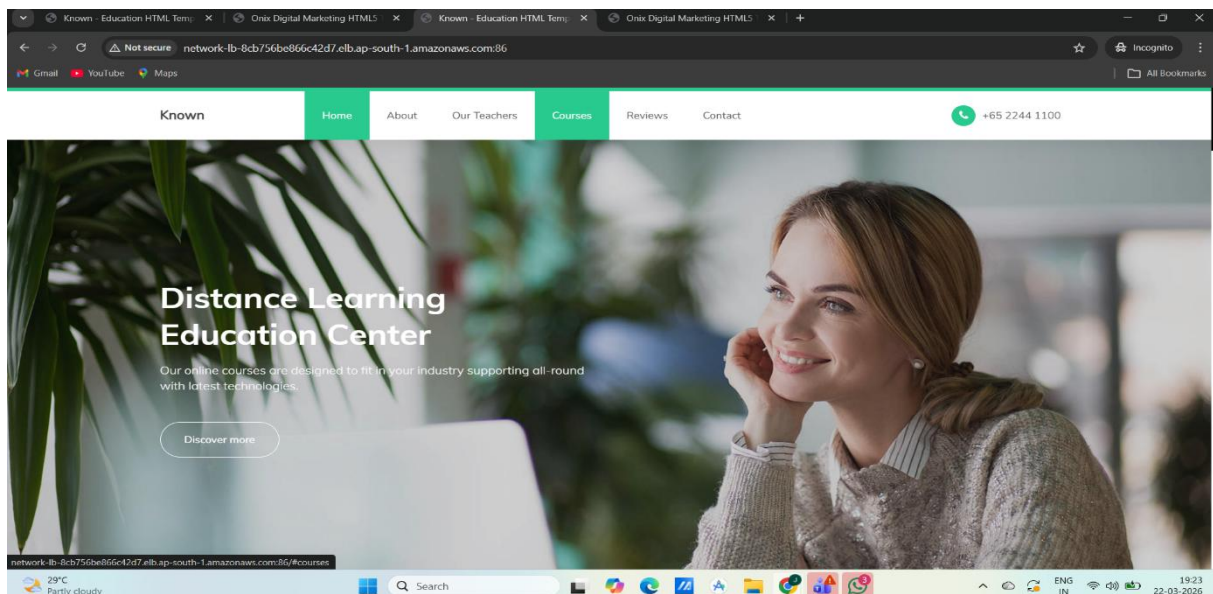
## Screenshot: Copy Load Balancer DNS Name

- Click on the Load Balancer name
- Copy the DNS name



## Screenshot: Access Application via Load Balancer (Port 86)

- Open a new browser tab ,Paste the Load Balancer DNS name
- Add :86 at the end of the URL
- Verify that the website opens successfully



## Screenshot: Access Application via Load Balancer (Port 92)

- Open a new browser tab
- Paste the Load Balancer DNS name
- Add :92 at the end of the URL
- Verify that the website opens successfully

